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Liquidity risk in banking: Integrating two hybrid Machine Learning models to introduce a comprehensive risk assessment system

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Abstract

The financial crisis highlighted liquidity issues, prompting studies on factors influencing liquidity risk in banking. Given its significance, this research intends to identify liquidity risk factors in banking using a regression model which is crucial for effective risk management by financial authorities. The results illustrate that Cost-To-Income and bank size are the most important variables which affect liquidity risk positively. Moreover, Return On Assets has also a significant negative impact and macroeconomic factors have insignificant negative effect on liquidity risk of banks. Prudently managing liquidity risk and establishing a Risk Assessment System (RAS) is crucial, as mishandling can lead to bank bankruptcy. A RAS is presented in this study to classify banks into four risk levels, utilizing four advanced machine learning models. Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN) and two hybrid models: Grid Search-Cross Validation Dropouts Meet Multiple Additive Regression Trees Extreme Gradient Boosting (GSCV- DART-XGBOOST) and Bayesian Random Forest Light Gradient Boost Machine (Bayesian-RF-LGBM) will be used in this research. Their results will be compared by 5 evaluation metrics to obtain the most accurate model. The dataset covers Iranian banks from 2013 to 2022, examining the pivotal role of effective liquidity risk management. The research findings confirm that GSCV-DART-XGBOOST is the most accurate method with the accuracy of 0.97 and can be applied in risk assessment of banks.

Keywords: Liquidity risk, Machine learning, Risk Assessment System, GSCV-DART-XGBOOST, Bayesian-RF-LGBM

1. Introduction

Banks have consistently held a decisive role in ensuring the smooth functioning of the economy. Across various economic systems, they play a primary role in executing financial policies. Its importance lies in the collection of deposits, provision of loans, creating liquidity and transforming risk (Mazreku, et al., 2019; Distinguin, et al., 2013). A key aspect is their ability to finance illiquid assets through liquid liabilities, especially vital when banks face the early liquidation of their assets to meet their financial obligations for several reasons, such as a bank rush (Berger & Bouwman, 2009). The importance of banking sector can be recognized and since banks are prone to bankruptcy if they do not manage their liquidity and the whole economy will suffer from its consequences, this research is dedicated to a comprehensive study of bank liquidity.

The term "liquidity" in financial markets encompasses various dimensions. Initially, liquidity represents the capability of a financial firm to maintain constantly an equilibrium between the financial inflow and outflow over time. Second, liquidity is a measure of the capability of a financial firm to turn an asset into cash quickly, without capital loss or interest penalty. Third, liquidity is interpreted as the capability of a bank to raise funds on the wholesale financial markets - first of all on the unsecured interbank market - by increasing its liabilities (Vento & La Ganga, 2009; BCBS, 2008).

Liquidity risk (LR), alongside credit risk, operational risk and market risk, is categorized as a financial risk. These risks have significant influence on how banks conduct their operations and banks will succeed only

if these risks are properly managed (Tavana, et al., 2018). Liquidity risk is defined as the risk that a financial firm, either does not have enough financial resources to meet its obligations as they fall due, or can obtain such funds only at excessive cost (Vento & La Ganga, 2009). Liquidity risk management involves a cost-benefit trade off, because a bank will be able to obtain funding provided and it is willing to pay the prevailing market prices, or has the choice of selling or committing its assets. An exploration of liquidity risk can lead to crucial results such as an understanding of the determinants variables of liquidity risk and specially being aware of the outcome of adopted policies in the future.

The Basel Committee (2006) differentiated the liquidity risk in funding liquidity risk defined as ‘the risk that the firm will not be able to efficiently meet expected and unexpected current and future cash flow without affecting daily operations or the financial condition of the firm’ and market liquidity risk characterized as ‘the risk that a firm cannot easily eliminate a position without significantly affecting the market price (Kumar & Chand Yadav, 2013; Haan & van den End, 2013). Therefore, from 2006 onwards, such distinction between funding liquidity risk and market liquidity risk became a prevalent feature in all literature on liquidity risk management.

Banks engaging in funds and maturity transformation forms the fundament of liquidity risk creation (Goodhart, 2008). The classic mismatch gap arising from the disparity in maturities between assets and liabilities introduces structural risk. Excessive liquidity can lead to inefficient resource allocation, while insufficient liquidity may reduce deposit interest rates, escalate debt, and ultimately result in bank failure (Matz & Neu, 2007). Banks hold substantial amount of short-term liabilities to finance long-term assets. Due to the nature of the short-term liability’s contracts, the depositors can claim their funds back at any time. Consequently, liquidity risk emerges on the liability side when depositors simultaneously withdraw funds while the bank is unable to fulfill its financial obligations (BCBS, 2008). Therefore, maintaining an optimal level of liquidity is crucial for banks to maximize profits and meet liabilities, ensuring resilience during various stress events (Zhang, et al., 2020).

To manage risk, banks commonly address withdrawals by either liquidating assets or securing funds from the money market (Chiorazzo, et al., 2018). While most assets can be converted into cash during the liquidation process, the sale of some assets at low prices can incur costs, potentially elevating insolvency risk. As a result, preparing a risk management strategy is necessary for all banks, achievable through an examination of the liquidity risk measure and its determinants. Effective liquidity risk management enhances a bank's ability to meet obligations promptly and diminishes the likelihood of adverse situations. The absence of risk management system to measure and assess liquidity risk in banks is a key factor that leads to unforeseen financial problems and bankruptcy (Cucinelli, 2013). This underscores the necessity for specific regulations addressing both the management and measurement of liquidity risk, aiming to enhance overall stability in the financial system (Jasienè, et al., 2012).

When certain banks take higher levels of risk, other banks may be encouraged to pursue similar strategies if their managers anticipate potential rescues in times of trouble. Hence, these risk-taking strategies may be mutually reinforcing in some circumstances (Bonfim & Kim, 2012). These collective risk-taking strategies may be detrimental for the whole economy and illustrates that examining liquidity risk is a key factor in preventing the possible crises in banking system.

Preceding the 2007–2009 financial crisis, liquidity risk was predominantly perceived as a factor influencing performance metrics. It highlighted the vulnerability of banks to liquidity risk and implied that the identification of suitable risk management for diverse business models was challenging (Galletta & Mazzù, 2019). The explosion of the crisis led to an interest in the issues of liquidity risk, including the factors determining this risk (Wójcik-Mazur & Szajt, 2015). Given the significance of this matter and the potential

for risk managers and bank authorities to manage liquidity risk with awareness of its determinants, this research aims to present the determinants of liquidity risk in the banking system, considering the factors applied in various studies. A regression model will be employed to estimate the relationship between selected variables and liquidity risk, focusing on identifying the most influential ones.

For the assessment of liquidity risk, the Basel Committee established minimum standards known as Basel III (2010). The initial tool for evaluating liquidity risk, the liquidity coverage ratio (LCR), was introduced in 2013 (BCBS, 2013), followed by the unveiling of the net stable funding ratio (NSFR) in 2014 (BCBS, 2014). This news had big impact on banks, because they were required to hold a level of capital and liquidity higher than in the past (Horvát, et al., 2012). The minimum liquidity standards are based on two mentioned complementary ratios. Although these ratios reflect the concept of liquidity risk correctly, implementing them in a banking system is not practical due to the complexity of calculations confidentiality of required data (Tavana, et al., 2018).

Thus, most researchers decide to choose measures to proxy bank liquidity, such as transaction deposits (Gatev & Strahan, 2006), the rate of changes in the aggregate balance sheet (Adrian & Shin, 2008), liquidity creation (Berger & Bouwman, 2009), and banks' bids in central banks' refinancing operation auctions (Drehmann & Nikolaou, 2013). Many published articles exploring methods for calculating liquidity risk have employed diverse ratios tailored to the specifics of their research structures. Consequently, this study aims to propose a unique ratio for evaluating liquidity risk, specifically designed to align with our selected methodology and research structure.

Most banks are facing problems due to poor liquidity risk management (BCBS, 2013). Thus, it is paramount to address liquidity risk prudently and establish a Risk Assessment System (RAS) by an efficient novel method. Since mismanagement of liquidity risk results in bank's bankruptcy, having awareness of each bank's risk level proves highly advantageous for banking authorities. Banks could be placed in four risk levels: high risk, medium risk, low risk and no risk. High risk banks maintain fewer liquid assets compared to medium and low risk ones. As the bank's managers get informed of the risk level, they will be able to monitor the outcomes of their policies and make prompt improvements when necessary. From macroeconomic perspective, due to the adverse effect of banks' bankruptcy on the whole economy, economic officials of countries can examine the general condition of banking system of their country and provide them a substantial insight. Therefore, we present a risk assessment model that classifies selected banks into four risk level groups using four novel classification machine learning models. The most accurate model will be chosen by comparing the results of applied methods.

The research is organized as follows: In section 2, first, several articles that carried out researches in the field of liquidity risk are reviewed. Then, the main factors affecting liquidity risk, their applied methods and their liquidity risk ratio are recognized by studying them. Section 3 introduces the models which are applied by this research and then presents the structure and methodology of the proposed models. The details of input variables and data will be discussed in the following of this section. The findings of models will be presented in section 4 and their results will be compared in order to introduce the most accurate model. Finally, in Section 5, the discussion of research, practical contribution and the proposes extracted from the research are expressed, respectively.

2. Effective variables on liquidity risk

Liquidity risk stands as a key concern in the bank's risk management strategies. Liquidity measures a financial firm's ability to function normally and maintain a balance between financial inflows and outflows over the period (Vento & La Ganga, 2009). Bank liquidity is a function of bank-specific and

macroeconomic factors. Bank-specific include factors such as non-performing loans, bank capital adequacy, bank deposits and profitability. While macro factors are external factors that affect the bank's liquidity but are not under the management of the bank but influence the economic and legal environment affecting the bank's functioning and banks liquidity position. Macroeconomic factors that may affect the bank's liquidity include factors such as Gross Domestic Product (GDP), interest rate (IR), inflation rate (INFR) and unemployment rate (UR) (Mazreku, et al., 2019).

It is crucial to emphasize that banks can manage liquidity effectively when the factors influencing this risk are accurately identified. This requires a precise internal understanding of the factors affecting liquidity and an assessment of how macroeconomic factors behave with regard to liquidity. By achieving consensus in both dimensions, many shortcomings in the current structure of bank liquidity can be addressed. Literature on the topic of bank liquidity determinants offers a range of studies that empirically validate the influence of microeconomic (bank specific) and macroeconomic factors over the liquidity of banks. In the following, some researches which have examined the effects of micro and macro factors on liquidity risk will be presented.

After the financial crisis, Berger & Bouwman (2009) applied four measures to estimate liquidity risk that differ in how off-balance sheet activities are treated and how loans are classified and applied them to data on virtually all U.S. banks from 1993 to 2003. Based on the preferred “cat-fat” liquidity creation measure, the relationship between capital and liquidity creation is positive and significant for large banks, insignificant for medium banks, and negative and significant for small banks. The research findings demonstrate that bank liquidity creation is positively correlated with bank value and only large banks could easily meet the liquidity needs, while small banks needed to reserve high liquidity. Moreover, deposits as the main source of bank liquidity should be increased to reduce liquidity shocks in banks.

Mugenyah (2015) conducted a study over 43 commercial banks licensed in Kenya between 2010 and 2014. He used multiple regression analysis to determine the effects of capital adequacy ratio, liquid assets ratio, ownership type, size, and leverage to identify the impact over liquidity risk. The study concluded that capital adequacy had a positive effect while liquid asset ratio, ownership type, size, and leverage negatively affected liquidity risk. The research findings confirm that capital adequacy ratio, liquid asset ratio, ownership type, size and leverage were significant determinants of liquidity risk.

Wójcik-Mazur & Szajt (2015) conducted a study over old European Union countries and so-called New European Union countries to measure the liquidity risk of bank-specific and macroeconomic factors. The study found internal determinants that affect the level of liquidity risk, regardless of the form of liquidity risk measure adopted and the country in which they operate. Macroeconomic factors such as GDP growth negatively correlate with liquidity in both new and old European Union countries. The increase of inflation level affects the increase of liquidity level.

Tran, et al (2016) investigated the relationship between liquidity creation, regulatory capital, and profitability using an unbalanced quarterly panel data of all U.S. banks from 1996 to 2013. The research findings indicate that legal capital investment and liquidity creation, after controlling for bank profitability, have a direct and mutual impact on each other. This relationship is primarily managed and controlled during non-crisis periods, predominantly by smaller banks. This depends on the level of legal capital in banks and how this measure of capital is assessed. The results show that banks with higher liquidity and a greater indication of liquidity shortage risk have lower profitability.

In another study, Abdul-Rahman, et al (2017) studied the impact of financial structure on bank liquidity risk. The data set of the Malaysian banking industry is selected to compare the financial liquidity risk relationship between 17 Islamic and 27 traditional banks from 1994 to 2014. There are 3 kinds of

independent variables in this research. Bank specific, macroeconomic and financial structure variables such as real estate financing. The results illustrate that increasing number of real estate financing and short-term stability of the Islamic banks may increase both their short- and long-term liquidity risks. On the other hand, even though real estate financing does not affect liquidity risks of the conventional banks, increasing short-term stability and financing specialization may increase their long-term liquidity risk.

Zaghdoudi & Hakimi (2017), covering ten Tunisian banks representing all Tunisian banking sectors, observed from 1980 to 2015. This paper's objective is to identify the key determinants of liquidity risk of these banks in order to manage this major risk to avoid both their liquidity drying up and their bankruptcy. The econometric results, based on panel data analysis, represent that the liquidity risk of Tunisian banks depends on the bank's internal factors such as loans and advances, bank size, etc. Among the macroeconomic factors, economic growth has a positive and significant effect, whereas inflation impacts negatively but not significantly the liquidity risk of Tunisian banks.

Chen, et al (2018) investigated the liquidity risk related to bank specific, government regulation and macroeconomic drivers by using an unbalanced annual panel dataset of commercial banks in 12 advanced economies over the period 1994–2006. They applied the fixed effects regression model to estimate the causes of liquidity risk. The results show that liquidity risk is an endogenous factor of bank performance. They realized banks with high levels of illiquid assets in loans may be compensated by higher net interest incomes. Moreover, a further macro key factor is geography. In fact, developing countries were shown to have a different liquidity risk cause to developed nations due to macroeconomic factors, like unemployment, education levels, and financial literacy.

Galletta & Mazzu (2019) carried out an analysis of liquidity risk drivers on a European sample of different banking types in the 2011–2017 time period. They applied dynamic panel regression model with clustered heteroscedasticity and standard errors and Generalized Method of Moments (GMM) dynamic panel estimator to examine the effects of variables on liquidity risk. They also introduced efficiency ratio, return on assets (ROA), net interest margin, risk-weighted assets, logarithm of total assets, capital Ratio equity, non-interest income/ total income and Non-deposit funding/ total liabilities as effective variables. They concluded that as banking size increases, risk also increases, whereas liquidity risk does not decrease with a bank's capital ratio.

Mazreku, et al (2019) applied a linear regression model using Pooled Ordinary Least Squares, a Fixed Effects Model, a Random Effects Model and a Hausman-Taylor regression model in order to estimate the relationship between liquidity risk and various effective variables on a set of data collected from banks in nine Balkan states, during the period 2000-2015. They considered non-performing loans, capital adequacy, etc. as bank specific and GDP, inflation, etc. as macroeconomic inputs. The research concluded that specific factors and macroeconomic factors, specifically, capital adequacy, non-performing loans, deposit growth, GDP, unemployment rate and marginal interest rate, significantly affect bank liquidity. However, inflation and profitability do not.

Jia, et al (2020) selected the four economic factors of unemployment, GDP, inflation, and interest rates and the noneconomic factor of management efficiency to test the correlation, and selected data from all commercial banks from 1996 to 2015, using multiple linear regression and normality test, linearity test and multicollinearity test to test the relationship between independent variables and dependent variables. Research shows that GDP and liquidity risk are negatively correlated. This means that a higher GDP growth rate may increase the liquidity risk of Malaysian commercial banks. At the same time, there is weak evidence to support the notion that inflation rate, unemployment rate, interest rate and management efficiency impact banks' liquidity.

Mohammad, et al (2020) selected 145 commercial banks during the period 1996- 2015 to discuss the liquidity risks faced by Islamic, commercial and hybrid banks. In addition, the macroeconomic and internal variables of the bank are also considered, and the regression model of random effects panel data is used to analyze the factors that affect bank liquidity risk. The findings for the sampled banks demonstrate that Islamic banks are more exposed to liquidity risk than conventional and hybrid banks. In addition, the results show that the stringency of capital regulations and credit risk have a negative and significant impact on liquidity risk. Moreover, the results demonstrate that liquid assets and long-term debts are positively associated with liquidity risk exposure.

Tran (2020) selected banks of America from 2001 to 2015 as a sample and examined the relationship between functional diversification and bank liquidity creation. They applied Cat-fat ratio, as a proxy of bank liquidity creation that was presented by Berger & Bouwman (2009) and considered some bank specific variables such as bank size, capital ratio, bank performance, etc. Various models were applied which include panel data regression models and Some econometric methods. Research shows that the impact of shifting to non-traditional activities on the creation of liquidity is more pronounced in large banks. Banks with insufficient liquidity creation, due to a higher proportion of non-interest income, are more likely to reduce liquidity creation.

Zhang & Zhao (2021) selected 16 commercial banks listed on Shanghai and Shenzhen A-shares as research samples and adopted data from 2010 to 2019. The study makes an empirical analysis on the factors affecting the liquidity risk of listed commercial banks. Firstly, the factors affecting liquidity risk of listed commercial banks are divided into external and internal levels. They considered GDP, Consumer Price Index (CPI) and Total money supply growth rate as the representative indicators of external environment. Besides, total assets, liquidity ratio, capital adequacy ratio, etc. as internal ones. Finally, they adopted Panel data and regression models to 16 listed commercial banks as a whole, 5 large state-owned listed commercial banks and 11 other medium and large listed commercial banks, through Hausman test and F test.

Mishra, et al (2021) attempted to forecast the financial distress of Indian commercial banks by developing an early warning model for bankruptcy prediction. The sample size for the study is 22 private, 12 public and 41 foreign banks. Logistic regression, linear discriminant analysis and artificial neural network models have been applied on the last 5 years' (2015–2019) data of these banks. The input variables for all three models were: long-term debt/ total debt, working capital/ total assets, quick ratio and growth in total assets. Data analysis results reveal the logistic and other models exhibiting similar prediction accuracy. The results of the ANN prediction model exhibit better prediction accuracy.

Yan & Song (2022) explored an intelligent identification method of influencing factors of commercial bank liquidity risk and established an early warning index system. They used 25 liquidity risk indicators as input variables. This paper uses web crawler technology and text analysis to realize the intelligent analysis of commercial bank liquidity risk keywords and uses grounded analysis technology to identify the risk factors that affect the deposit-loan maturity mismatch. They successfully designed a model with the research time span covering the years 2000 to 2020 and the geographical scope involving selected samples from commercial banks in China. The results show that the constructed model has high accuracy, which can provide support for banks and relevant government departments to formulate and resolve bank liquidity risk.

According to the review of researches that have examined the liquidity risk, we can come to the following conclusions: Initially, there is a vast range of variables which authors decide to select based on the structure of their studied area. As mentioned above, mainly, macroeconomic and bank specific factors affect liquidity risk fluctuation. Table 1 illustrates a summary of effective variables adopted by various researches. The

most important factors can be conceived by table 1. Besides, another issue which can be comprehended is the ratio that authors have considered for liquidity risk. It can be understood that there are different ratios which present liquidity risk based on the selected definition. Table 2 summarizes various adopted ratios in the recent articles. The next point is the importance of applied methods. Most of the articles applied only statistical methods such as econometric and regression models in order to examine the relationship between factors and liquidity risk. However, there are few researches that used machine learning for various purposes. A comparison of applied methods in the liquidity risk researches will be illustrated by table 3.

Table 1 Input variables in different published papers

Research	Effective Variables
Berger & Bouwman (2009)	Bank specific: Capital adequacy, Bank risk, Bank size, Bank holding status, Mergers, Local market competition, Local market environment
Giannotti, et al (2011)	Bank specific: Logarithm of total assets, Euro Index, Euro Interbank rate, Lending facilities, Bank type, Short-term lending, Medium- to long-term lending
Akhtar, et al (2011)	Bank specific: Return On Equity (ROE), Return On Asset (ROA), Bank size, Capital adequacy, Net-working capital
Iqbal (2012)	Bank specific: Bank size, Non-performing loan ratio, ROA, ROE, Capital adequacy
Munteanu (2012)	Bank specific: Asset quality, Interbank funding, Funding cost, Capital adequacy, Cost-to-income, Credit risk Macroeconomic: Interest rate, GDP growth, Inflation rate, Unemployment
Choon, et al (2013)	Bank specific: Bank size, Capital adequacy, Profitability, Non-performing loans Macroeconomic: GDP, interest rate, Financial crisis
Jedidia & Hamza (2015)	Bank specific: Bank size, Capital adequacy, ROA, Investment assets Macroeconomic: GDP growth, Financial crisis, Foreign or domestic banks
Trenca, et al (2015)	Bank specific: Logarithm of total assets, ROE, Loan losses/ Net loans, Share of shareholders' equity /Total assets Macroeconomic: GDP growth, Inflation rate, Unemployment rate, Net lending /Net borrowing
Rahman & Banna (2015)	Bank specific: Bank size, Net-working capital, ROE, Capital adequacy, ROA
Tran, et al (2016)	Bank specific: Bank risk, Bank size, Productivity growth, Operating management Macroeconomic: GDP growth Market: Market concentration, Stock market volatility
Zaghoudi & Hakimi (2017)	Bank specific: Credit risk, Bank size, Capital adequacy, Herfindahl-Hirschman Index Macroeconomic: GDP, Inflation rate, Financial crisis
Galletta & Mazzu (2019)	Bank specific: Efficiency, ROA, Net interest margin, Risk-weighted assets, Logarithm of total assets, Capital Ratio equity, Non-interest income/ total income, Non-deposit funding/ total liabilities
İncekara & Cetinkaya (2019)	Bank specific: Bank size, Non-performing loans, ROE, Capital adequacy, ROA, Liquid assets Macroeconomic: GDP, CPI
Mazreku, et al (2019)	Bank specific: Non-performing loans, Capital adequacy, Deposits, Profitability, size Macroeconomic: GDP, Inflation, Interest rate, Unemployment
Al-Duhaidahawi, et al (2020)	Bank specific: Liquidity ratio, Current ratio, Loan-to-Deposit ratio
Duan & Niu (2020)	Bank specific: Credit risk, ROA, ROE, Bank size, Capital ratio Macroeconomic: Interest rate, Inflation, GDP, Bank concentration, Bank efficiency
Jia, et al (2020)	Macroeconomics: GDP, Inflation, Interest Rate, Unemployment Non-Economic: Management efficiency
Tran (2020)	Bank-specific: Bank size, Capital ratio, Bank performance, Asset growth, Earnings volatility, Loan portfolio, Bank risk, Non-interest income/net operating income Macroeconomic: GDP, CPI, Total money supply
Zhang & Zhao (2021)	Bank specific: Total assets, Liquidity ratio, Capital adequacy, Cost-to-income, ROE, Non-performing loan
Yan & Song (2022)	25 variables including bank specific, policy environmental, industry environmental, institutional environmental factors

Considering the mentioned points, in this research, according to the effectiveness of variables on the selected data, we will benefit from both types of variables. Since the most vital macroeconomic variables have been GDP, inflation rate, interest rate and unemployment rate in various studies, this research has chosen them as macroeconomic factors. With regard to bank-specific ones, 5 factors have been selected which include Bank size, Capital ratio, Cost-To-Income ratio, Return On Asset and Return On Equity. Due to the findings of recent articles and the interpretation of authors, selected inputs have possibly the most significant impact on the liquidity risk of banking system. In the next sections, the level of effectiveness of selected variables will be inquired.

2.1. Selected ratios in respect of liquidity risk

As it was discussed earlier, although the ratios introduced by Basel III (2013; 2014) are able to present the concept of liquidity risk properly, it is not practical to apply them for examining liquidity risk in a banking system due to the fact that these ratios demand data which are confidential and are not accessible to public. Besides, they include complex calculations in order to obtain considered coefficients. As a result, most of the researches carried out in the field of liquidity risk have decided to estimate indirect measures to calculate bank liquidity based on the structure of their research.

Table 2 *Liquidity risk ratios in different published papers*

Researches	Liquidity Risk ratios							
	LCR	NSFR	Loan/ Deposit	Cat- Fat	Cash/ Asset	Asset/ Liabilities	Loan/ Asset	Financial Gap
Berger & Bouwman (2009)				✓				
Akhtar, et al (2011)					✓			
Iqbal (2012)					✓			
Munteanu (2012)							✓	
Choon, et al (2013)			✓					
Ben-Moussa (2015)			✓		✓			
Jedidia & Hamza (2015)					✓		✓	
Trenca, et al (2015)			✓					
Abdul-Rahman, et al (2017)	✓	✓						
Chen, et al (2018)								✓
Tavana, et al (2018)						✓		
İncekara & Cetinkaya (2019)								✓
Mazreku, et al (2019)						✓		
Al-Duhaidahawi, et al (2020)						✓		
Duan & Niu (2020)				✓				
Jia, et al (2020)			✓					
Mohammad, et al (2020)								✓
Tran (2020)				✓				
Zhang & Zhao (2021)						✓		
Mishra, et al (2021)						✓		
Yan & Song (2022)	✓							

It was mentioned that one important issue which can be comprehended by studying carried out researches in the field of liquidity risk is the ratio that authors have considered for liquidity risk. Table 2 summarizes various adopted ratios in the recent articles. It can be understood that there are different ratios which present liquidity risk based on the selected definition. As discussed above, authors are able to choose a ratio in order to calculate liquidity risk considering the type of data and issue which they are investigating. Therefore, in this research, we have decided that the ratio of current assets to current liabilities, known as current ratio (CR), can present liquidity risk better than any other one. Since the mentioned ratio has been

selected by authors several times, the decision of authors is probably authentic and will have the most accurate results. The current ratio indicates the responsibilities of bank versus its liabilities and must take a normal value of at least 1. Thus, when the current ratio starts to fall from this amount, the risk begins to unfold. Due to the definition of liquidity risk, liquidity risk can be formalized using the following function:

$$Liquidity Risk = \begin{cases} 1 - CR & \text{if Current Ratio} < 1 \\ 0 & \text{if Current Ratio} \geq 1 \end{cases} \quad (1)$$

2.2. Applied methods to examine liquidity risk

This research intends to apply methods which possess significant accuracy and have not been considered in recent researches investigating liquidity risk. Machine learning includes novel models with great precision and validation that have been introduced in numerous fields. However, the authors noticed that liquidity risk have not been measured and examined by machine learning but few articles. Testing the ability of machine learning in this specific field is of substantial importance and can bring interesting results.

Table 3 Methods applied in different published papers

Researches	Methods	
	Statistical Analysis	Machine Learning Methods
Berger & Bouwman (2009)	✓	
Giannotti, et al (2011)	✓	
Akhtar, et al (2011)	✓	
Iqbal (2012)	✓	
Munteanu (2012)	✓	
Choon, et al (2013)	✓	
Jedidia & Hamza (2015)	✓	
Rahman & Banna (2015)	✓	
Tran, et al (2016)	✓	✓
Zaghdoudi & Hakimi (2017)	✓	
Chen, et al (2018)	✓	
Tavana, et al (2018)		✓
İncekara & Cetinkaya (2019)	✓	
Al-Duhaidahawi, et al (2020)		✓
Duan & Niu (2020)	✓	
Jia, et al (2020)	✓	
Mohammad, et al (2020)	✓	
Tran (2020)	✓	
Zhang & Zhao (2021)	✓	
Mishra, et al (2021)	✓	✓
Yan & Song (2022)		✓

As table 3 illustrates, the majority of authors which examine liquidity risk, have used statistical methods, econometric and regression for instance, in their researches. Similarly, we have chosen regression model to inquire the effectiveness of selected variables on liquidity risk. Although machine learning models provide spectacular accuracy in their tasks, they are not applied in liquidity risk issues. Thus, it is of great significance to use novel machine learning methods to examine liquidity risk. we aim to use four different machine learning models in this research which classify chosen banks into four different risk levels. The models include: Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN) and two hybrid models which are Grid Search-Cross Validation Dropouts Meet Multiple Additive Regression Trees Extreme Gradient Boosting (GSCV-DART-XGBOOST) and Bayesian Random Forest Light Gradient Boost Machine (Bayesian-RF-LGBM). Their results will be compared to obtain the most accurate model.

3. Methodology

In this section, we will examine the mathematical concept of the models used in this research. Neural networks and machine learning models have been used in many important issues due to their powerful

features such as the ability to solve nonlinear problems, high learning power, teachability of past data, high accuracy and fault tolerance (Safiyari, et al., 2022). By contrast, these methods have not been used by most of the researches which aimed to examine liquidity risk during recent decade. As a result, the need to study liquidity risk of banks applying various machine learning models can be comprehended in order to estimate the accuracy and ability of machine learning in the field of liquidity risk. In this article, Regression method will be applied to investigate the relationship between input variables and liquidity risk. Two deep learning methods, CNN and LSTM, and two hybrid models GSCV-DART-XGBOOST and Bayesian-RF-LGBM, which benefit from an optimization algorithm to increase their accuracy, will be compared by their ability to classify selected banks based on their liquidity risk level and the best model will be presented. All mentioned models will be interpreted in depth.

3.1. Regression

Multiple regression analysis allows us to verify the presence of connections between several variables: dependent and multiple independent variables. In order to investigate the association between the liquidity risk, as the dependent variable, and selected independent variables, this study develops the following regression model:

$$LR_{it} = \beta_0 + \beta_1 SIZE_{it} + \beta_2 GDP_t + \beta_3 INF_t + \beta_4 UR_t + \beta_5 IR_{it} + \beta_6 CR_{it} + \beta_7 CTI_{it} + \beta_8 ROA_{it} + \beta_9 ROE_{it} + \varepsilon_{it} \quad (2)$$

Where, LR is the dependent variable, which indicates liquidity risk of i_{th} bank in year t. The $\beta_0, \dots, \beta_n$ are the regression coefficients. $SIZE_{it}$ indicates the dimension of i_{th} bank in year t. GDP_t , INF_t , UR_t and IR_{it} represent the annual change of GDP in year t, the annual change of consumer price index in year t, the annual change of unemployment change in year t and the lending interest rate of i_{th} bank in year t, respectively. CR_{it} is the ratio of total equity of i_{th} bank in year t to total assets of i_{th} bank in year t. CTI_{it} measures the ratio between i_{th} bank total expenses in year t and i_{th} bank total incomes in year t. ROA_{it} and ROE_{it} are the proxies of i_{th} bank returns in year t, measuring the return on assets and return on equity, respectively. ε_{it} is also the regression error term.

3.2. Classification

As the relationship between liquidity risk and its determinants is unfold, we aim to classify banks into four different liquidity risk levels including: high-risk, medium-risk, low-risk and no-risk. Clearly, this research intends to present a risk assessment system which enables the authorities to be aware of a countries' (even a region) banking system condition in case of its liquidity risk. As mentioned before, the equation which measures liquidity risk in this research is:

$$Liquidity Risk = \begin{cases} 1 - CR & \text{if Current Ratio} < 1 \\ 0 & \text{if Current Ratio} \geq 1 \end{cases} \quad (3)$$

Therefore, when a bank owns current assets as much as its current liabilities or more has no liquidity risk since it will be able to pay all the liabilities. By contrast, if the bank does not possess sufficient current assets, the bank will suffer from liquidity risk issues. In this research, we have divided the level of banks' liquidity risk into four categories based on the percent of their liquidity risk which can be interpreted as following equation:

$$LIQUIDITY RISK LEVEL = \begin{cases} NO RISK, & LR = 0 \\ LOW RISK, & 0\% < LR < 20\% \\ MEDIUM RISK, & 20\% \leq LR < 40\% \\ HIGH RISK, & LR \geq 40\% \end{cases} \quad (4)$$

We will benefit from four outstanding machine learning models for classification which have presented substantial accuracy in various classification researches. However, they have never been used in liquidity risk classifications. In the following, the details of them will be discussed.

3.2.1. GSCV-DART-XGBOOST

XGBOOST is an optimized distributed gradient boosting model. It is improved from the gradient boosting tree, which integrates several weak classifiers into a strong classifier, and makes predictions through the classification and regression tree model. Figure 1 presents the structure of XGBOOST model. The ensemble model of the tree looks like this:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (5)$$

In the formula: $\hat{y}_i$ is the predicted value of the i_{th} sample, $f_k(0)$ is the correlation between the structure of the k_{th} tree and the leaf weight, x_i is the input value of the i_{th} data point eigenvector, K is the total number of trees. The XGBOOST loss function L_{loss} is:

$$L_{loss} = \sum_{i=1}^n \varphi(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (6)$$

$$\Omega(f) = \gamma_T + \frac{1}{2} \lambda \|w\|^2 \quad (7)$$

In the formula: the first part of the gradient boosting algorithm loss is the training error of the predicted value $\hat{y}_i$ and the real value of y_i , the latter part is the regularization term, γ and λ represent the penalty coefficient for the model, T represents the number of leaf nodes, w represents the weight of the leaf. In order to reduce the loss function as much as possible, an incremental function can be added in each iteration. The objective function for t iterations can be written as:

$$L_{loss}^t = \sum_{i=1}^n \varphi(y_i, \hat{y}_i^{t-1} + f_t(x_i)) + \Omega(f_t) \quad (8)$$

Perform a second-order Taylor expansion for the above equation:

$$h_i = \partial_{y_i} \varphi(y_i, \hat{y}_i^{t-1}) \quad (9)$$

$$e_i = \partial_{y_i}^2 \varphi(y_i, \hat{y}_i^{t-1}) \quad (10)$$

In the formula: h_i and e_i are the first-order and second-order derivatives of the loss function. Define the sample set in each leaf of the m_{th} tree as $l_m = \{i | q(x_i) = m\}$. From this, the approximate objective function obtained by the second-order Taylor expansion of formula can be obtained:

$$L_{loss}^t = \sum_{j=1}^T \left[(\sum h_i) w_j + \frac{1}{2} (\sum e_i + \lambda) w_j^2 \right] + \gamma T \quad (11)$$

Define $G_j = \sum g_i$, $H_j = \sum h_i$ and then take the partial derivative of w and bring it into the objective function to get (Zhengzhong, et al., 2022):

$$L_{loss}^t = -\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{h_j + \lambda} + \gamma T \quad (12)$$

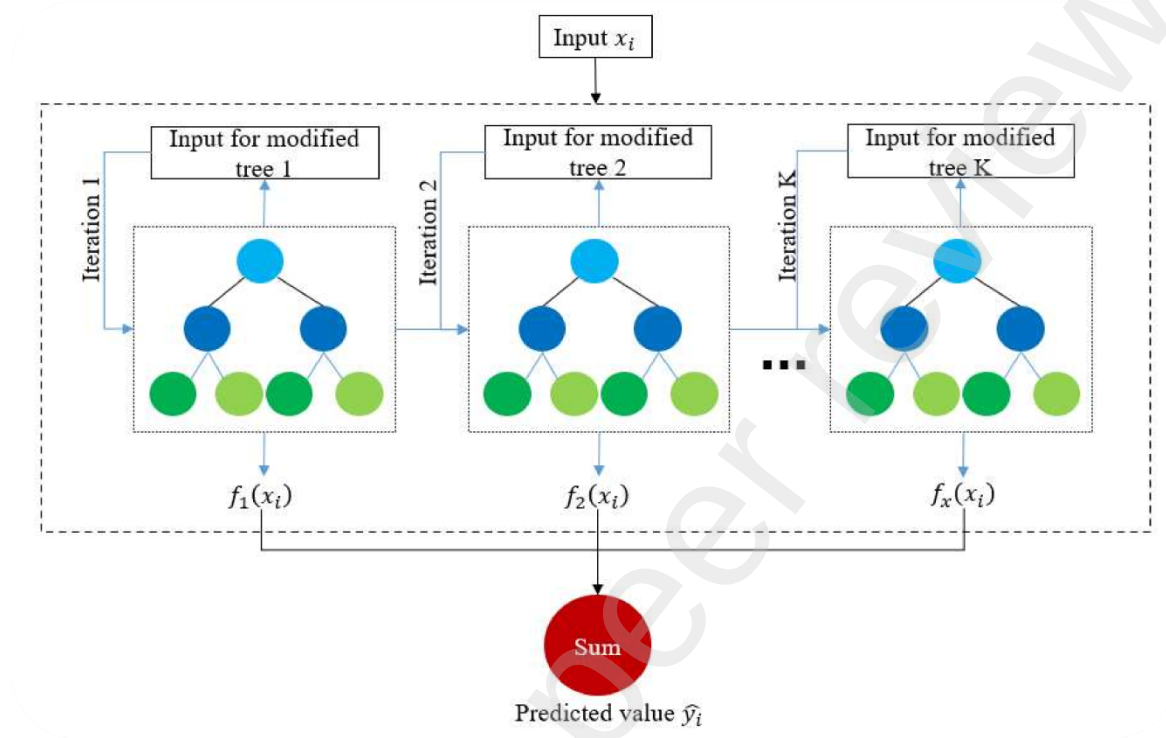


Figure 1 Architecture of an XGBOOST model

The XGBOOST classifier supports different types of boosting algorithms in the training process. One of the most important boosters is DART (Dropouts Meet Multiple Additive Regression Trees). DART is an enhancement of the tree boosting algorithm that introduces dropouts during the training of trees. It aims to improve regularization and robustness.

The accuracy of models can be improved by using the Grid Search (GS) and K-CV. The GS was developed to optimize complicated problems. The value of hyper-parameters was limited in a wide range instead of a default value. Within the specified hyper-parameters range, the model parameters are adjusted according to the step length. The hyper-parameter was set as the minimum value and added the value of step length in sequence until larger than the maximum value in the specified range. Then, the learners can be trained with the adjusted parameters to seek the best combination of hyper-parameters with the highest precision on the validation set. The GS is typically used with K-CV, which can produce the evaluation index for the model. When the model is trained, the database is split into two parts: a training set and a test set. K-CV is a commonly used method for training sets. It divides the training set into equal-sized parts. Each part of the training set is regarded as the validation set, and the remaining $K-1$ parts are considered as new training sets. Then, K models will be established, trained, and validated with $K-1$ training sets and K validation sets.

The accuracy of K models can be obtained, and the performance of the K-CV model is evaluated based on the average accuracy of the K models. Next, the parameters of the model are changed based on the GS, and the accuracy of model is recalculated. The accuracies of models with all parameter combinations are compared to determine the optimal parameter combination. Finally, the hyper-parameters were adjusted and set pairs within the specified parameters range in the primary learners, and the model with the highest accuracy was selected by K-CV and applied in prediction (Yan, et al., 2022). In this research, we apply GSCV to XGBOOST model to optimize model's hyper-parameters.

3.2.2. Bayesian-RF-LGBM

Light Gradient Boost Machine (LGBM) is a tree-based learning model. It combines multiple weak learners to improve the performance of the system and has a significant performance in terms of computation time and memory usage. Figure 2 shows the architecture of LGBM. In addition, LGBM uses two approaches: bagging and boosting. A decision step is sustained by the bagging approach depending on the majority votes of classifiers, whereas the boosting approach constructs classifiers to avoid errors for the trained classifier (Kilic, et al., 2023). The following equation represents the LGBM:

$$f(x) = \sum_{n=1}^N f_n(x) \quad (13)$$

where N trees for obtaining probable model $(f_1, f_2, \dots, f_n)$.

Random Forest (RF) is a decision tree algorithm that works through building several trees and can be described as a forest of trees in which each tree casts a vote for the most popular input vector class. A pool of individual tree arranged classifiers can be represented as following:

$$\{RF(y, a_n), n = 1, 2, \dots, i, \dots\} \quad (14)$$

where, RF is the classifier, $\{a_n\}$ stands for identically independent distributed random vectors, and each tree has a vote for most famous class at input variable y . The nature and dimension of a hinge on its use in tree building.

The building of each of the trees that form the forest, is the key to the achievement of the RF. The diversity of RF could be obtained by sampling from the feature set, from the data set, or simply by randomly varying some decision tree parameters. Because of its high execution speed, RF is an appealing classifier, and it typically has much higher classification efficiency compared to a single decision tree. Mostly, the higher the number of trees in the forest, the more strongly it appears (Hassan, et al., 2022).

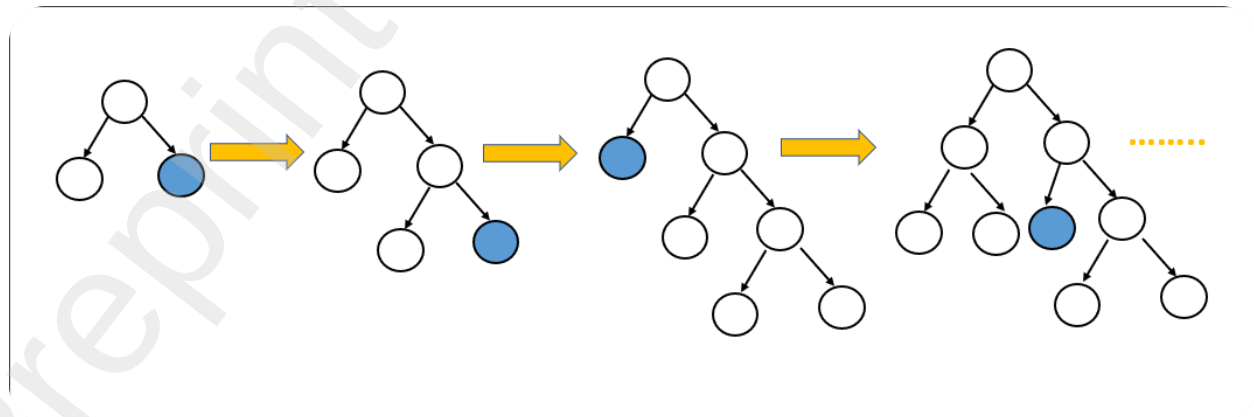


Figure 2 Architecture of an LGBM model

The Bayesian Optimization Algorithm can find the global optimum of $f(x)$ based on the Gaussian process regression (GPR) and acquisition function (AF). The GPR can obtain a posterior probability distribution from existing data observations, whereas AF uses a surrogate model derived from GPR to select the next point for evaluation. AF includes three main types: expected improvement (EI), probability of improvement (PI), and upper confidence bound (UCB), which are meant to balance the exploration and exploitation processes (Asante-Okyere, et al., 2022). Assume the point where the acquisition function $f(x)$ attains its maximum value at the current interaction x^+ . The probability improvement (PI) function can be expressed as follows:

$$PI(x) = P(f(x) \geq f(x^+)) = \phi(z) \quad (15)$$

Where $z = \frac{\mu(x) - f(x^+)}{\sigma(x)}$, ϕ is the cumulation distribution function of the normal distribution. The variance and the expectation of the prediction are represented as $\sigma(x)$ and $\mu(x)$. The expected improvement (EI) function is presented as:

$$EI(x) = \begin{cases} (\mu(x) - f(x^+))\phi(z) + \sigma(x)\varphi(z), & \text{if } \sigma(x) > 0 \\ 0, & \text{if } \sigma(x) = 0 \end{cases} \quad (16)$$

where φ is the function of normal distribution of probability density. The upper and lower confidence bond is given by following equations:

$$UCB(x) = \mu(x) + \beta\sigma(x) \quad (17)$$

$$LCB(x) = \mu(x) - \beta\sigma(x) \quad (18)$$

where $UCB(x)$ and $LCB(x)$ represent the upper and lower confidence bond functions, respectively, and β is the parameter that balances the exploration and exploitation capabilities of the acquisition function (Zhang, et al., 2020). In this study, we optimize the LGBM hyper-parameters by Bayesian algorithm which help us to determine the most suitable measures. Moreover, the type of boosting trees in LGBM can be carried out by different algorithm which the default one is traditional Gradient Boosting Decision Tree. However, we decided to consider Random Forest for boosting in order to increase its accuracy.

3.2.3. LSTM

Hochreiter & Schmidhuber (1997) have proposed LSTM to overcome the vanishing and exploding gradients problem related to RNN. The memory of LSTM cell will be stored and converted from input to output in cell state. LSTM cell is comprised of the forget gate, output gate, input gate and update gates. As the name indicates, forget gate decides what to forget from the previous memory units, the input gate decides what to accept into the neuron, the update gate updates the cell, and the output gate generates the new long-term memory. These four main components of LSTM will work and interact in a special manner, as it accepts long-term memory, short-term memory, input sequence at a given time step and generates new long-term memory, new short-term memory and output sequence at a given time step (ArunKumar, et al., 2021). The input gate decides which information must be transferred to the cell; the input gate is mathematically represented as following:

$$i_t = \sigma(W_i * [h_{t-1}, x_t] + b_i) \quad (19)$$

The operator '*' represents the element-wise multiplication of the vectors. The information to be neglected from the previous memory is controlled by forget gate which is mathematically defined as following:

$$f_t = \sigma(W_f * [h_{t-1}, x_t] + b_f) \quad (20)$$

The cell state is updated by the update gate, expressed mathematically by the following equations:

$$\tilde{c}_t = \tanh(W_c * [h_{t-1}, x_t] + b_c) \quad (21)$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t \quad (22)$$

The hidden layer of the previous time step is updated by the output gate which is also responsible for the updating the output as it is given by:

$$o_t = \sigma(W_o * [h_{t-1}, x_t] + b_o) \quad (23)$$

$$h_t = o_t * \tanh(c_t) \quad (24)$$

Figure 3 illustrates the structure of LSTM.

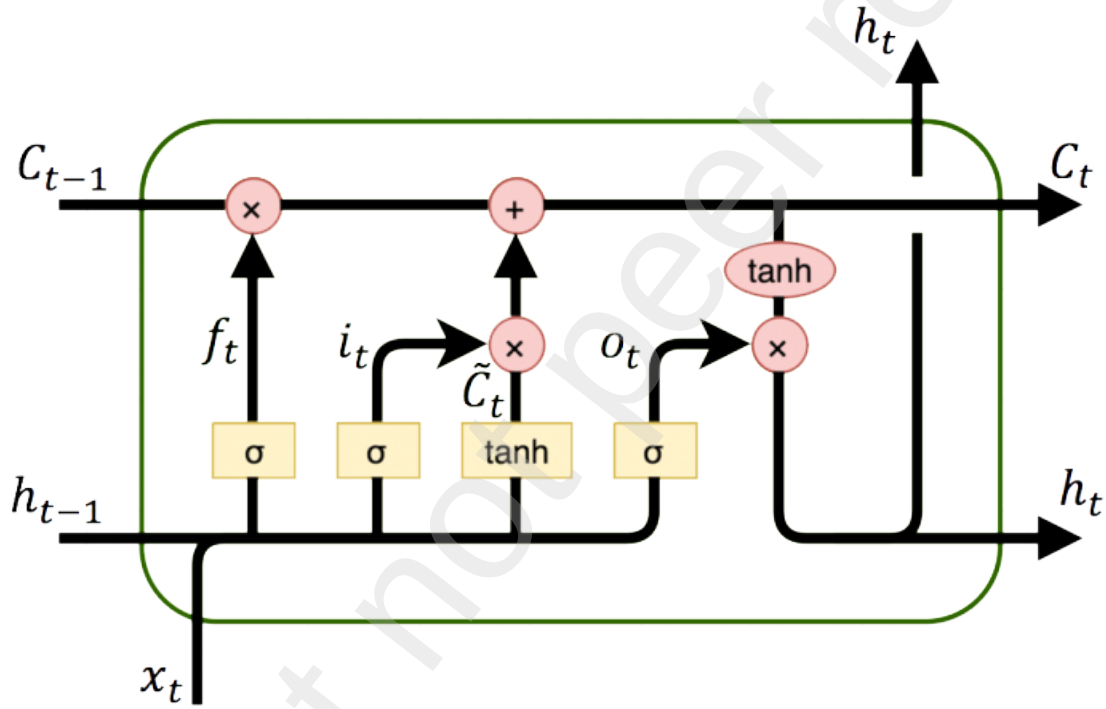


Figure 3 Architecture of an LSTM model

3.2.4. CNN

A mathematical model for CNNs can be characterized by means of the following equations and notations. Let $x \in R(N \times M)$ represent the input data, where N is the number of samples and M is the number of channels. Let $W^l \in R^{K \times K \times D^{l-1} \times D^l}$ denote the weight matrix of the l_{th} convolutional layer, where K is the filter size, D^{l-1} is the input channel, and D^l is the output channels. The convolution operation for the l_{th} layer can be written as:

$$z^l = f \sum_{i=1}^{D^{l-1}} (x * w_i^l) + b^l \quad (25)$$

where ‘*’ represents the convolution operation, f is the activation function, and b^l is the bias term for the l_{th} layer. The pooling operation, typically max pooling, down-samples the feature maps to reduce dimensionality and extract key features. Let p^l denote the pooling operation for the l_{th} layer, and p_{size}^l be the size of the pooling window. The output of the pooling operation can be expressed as:

$$\hat{x}^l = p^l(z^l, p_{size}^l) \quad (26)$$

$W_{F_{c,l-1}}$ represent the weight matrix for the l_{th} fully connected layer, where $H_{F_{c,l-1}}$ and $H_{F_{c,l}}$ are the sizes of the input and output layers, respectively. The output of the l_{th} fully connected layer is computed as:

$$Z_{F_{c,l}} = W_{F_{c,l}} * x_{F_{c,l-1}} + b_{F_{c,l}} \quad (27)$$

where $x_{F_{c,l-1}}$ represents the input to the l_{th} fully connected layer, $b_{F_{c,l}}$ is the bias term, and f is the activation function. The predicted output for a given input sample is given by:

$$\hat{y} = \sigma((Z_{F_{c,L}})) \quad (28)$$

where L is the index of the last fully connected layer and σ is the sigmoid activation function. The selection of loss function depends on the precise problem and can include cross-entropy loss, mean squared error, etc. $L(y, \hat{y})$ represent the loss function that measures the discrepancy between the predicted output $\hat{y}$ and the true label y . The CNN model is trained by minimizing the loss function using techniques such as gradient descent or its variants. The weights and biases are updated iteratively by computing the gradients of the loss function with respect to these parameters and adjusting them accordingly (Kumar, et al., 2024). Figure 4 indicates the architecture of CNN network.

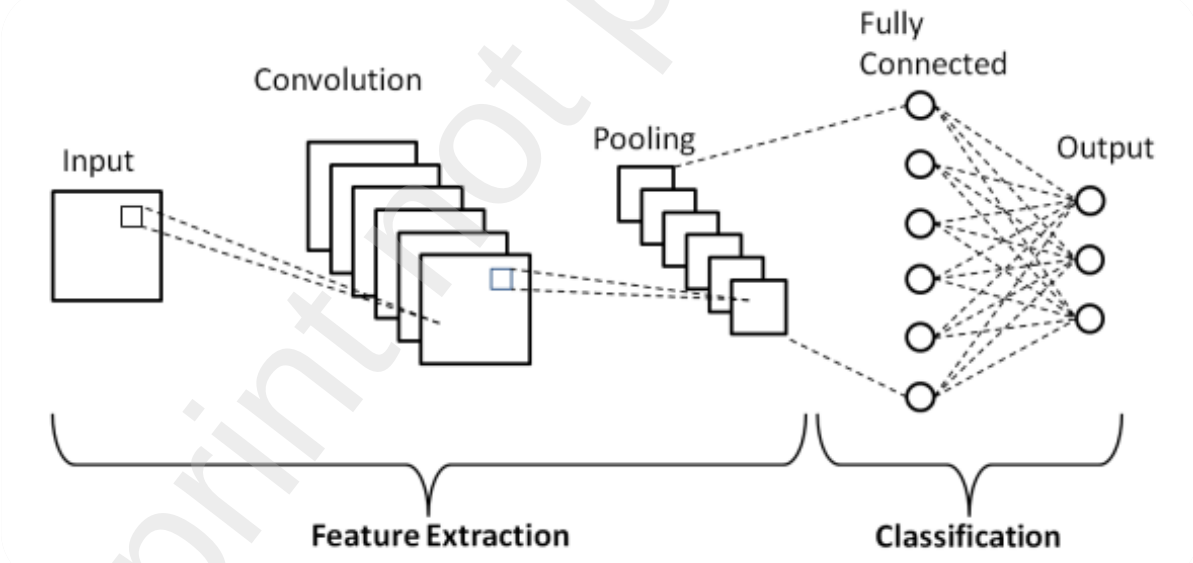


Figure 4 Architecture of an CNN model

3.3. Input variables

Liquidity risk can be affected by several various concepts. Some of them are related to banking system and are able to influence the structure of liquidity risk directly. The other category is variables that are important to the whole economy. They can change the policies taken by bank risk managers and the performance of banks indirectly. In this research, due to the studies carried out in section 2 which led to identification of

important variables and the characteristics of our selected data, a combination of bank-specific and macroeconomic variables has been selected as model's independent variables. The chosen factors are as follows: GDP, Inflation rate, Interest rate, Unemployment rate, Bank size, Capital ratio, Cost-To-Income ratio, Return On Asset and Return On Equity. As it was mentioned, the liquidity risk which this study aims to investigate by the ratio of current assets to current liabilities is the dependent variable of model.

The Return On Equity and the Return On Assets reflect the profitability of the bank both related to the assets and equity. ROA reflects a bank's management's capability to generate profits from the bank's assets. ROA is measured by the ratio of net profit to total assets. Similarly, ROE designates the return to shareholders on their equity, reflecting the bank's proficiency in using its own funds to generate profits. ROE is calculated by the ratio of net profit to total equity.

The natural logarithm of a bank's total assets is the proxy for bank size, influencing liquidity levels by affecting its ability to attract funds from various sources and the associated costs. Larger banks possess an intrinsic ability to attract numerous deposits with greater ease, enabling them to extend more loans at any given time. To measure operational efficiency, cost-to-income ratio is applied in researches. The ratio of total expenses to total incomes illustrates this index. Capital ratio is a proxy of a bank's capital situation. It is obtained by the ratio of total equity to total assets.

Inflation rate is determined by percentage change in the value of Consumer Price Index. It measures the change in the prices of a basket of goods and services in a year. As inflation grows, banks tend to hold more illiquid assets and lend more loans to earn higher interest incomes. It decreases the real rate of return not only on money terms but also on assets in general. Consequently, depositors tend to invest in illiquid assets instead of depositing in bank accounts to prevent from the depreciation of their money. Banks attract fewer deposits and banks' financing gaps increase which increases the vulnerability of banks to nominal values of loans provided to customers.

GDP is a proxy of total economic activity. It is commonly used as an indicator of the economic health of a country. For banks, GDP can be a key indicator to measure the demand for banking services in the context of receiving deposits and providing financing. For example, demand for differentiated financial products is higher during an economic boom and can improve the bank's ability to expand loan and securities portfolios to a higher level, improving the quality of their assets. Similarly, during the economic downturn, we have a reduction in a banks' credit facility. GDP is measured by the percentage change in GDP at constant prices. Since real GDP is adjusted for inflation, controlling for real GDP is meaningful and relevant to liquidity risk, as it indicates the changes in physical production. Interest rate as a measure of lending profitability at market level is calculated by the rates which banks determine on their loans. Higher interest rates could be naively interpreted as an explanation for lower bank liquidity. Level of unemployment, which is connected with demand for loans is measured by percent change of unemployment rate.

3.4. Data

The data comprises 10 categories of variables, including six bank-specific (Current ratio, Bank size, Capital ratio, Cost-To-Income ratio, ROA and ROE) and four macroeconomic (GDP, Inflation rate, Unemployment rate and Interest rate). The mentioned samples are obtained annually for 15 Iranian conventional banks covering from the beginning of 2013 to the end of 2022. The banks include Middle-East, Pasargad, Mellat, Tejarat, Sina, Saderat, Karafarin, Eqtesad-Novin, Parsian, Tourism, Melal, Resalat, Saman, Post and Refah. The dataset was gathered from the Central Bank of Iran (cbi, 2023), Securities and Exchange Organization (codal, 2023) and the bank's websites. There are 141 Observations in the sample. In order to classify Iranian banks, 105 data of each variable have been applied for model training and 36 data have been considered for predicting in the proposed models.

Figure 5 illustrates the number of banks in each liquidity risk category which have been chosen in our data. It can be realized that most of the banks were categorized as No Risk during recent years which indicates that Iranian banks do not take much risk due to the adverse economic situation.

4. Results and findings

In this section, the assessment of banks' risk is carried out by classifying them into four different categories

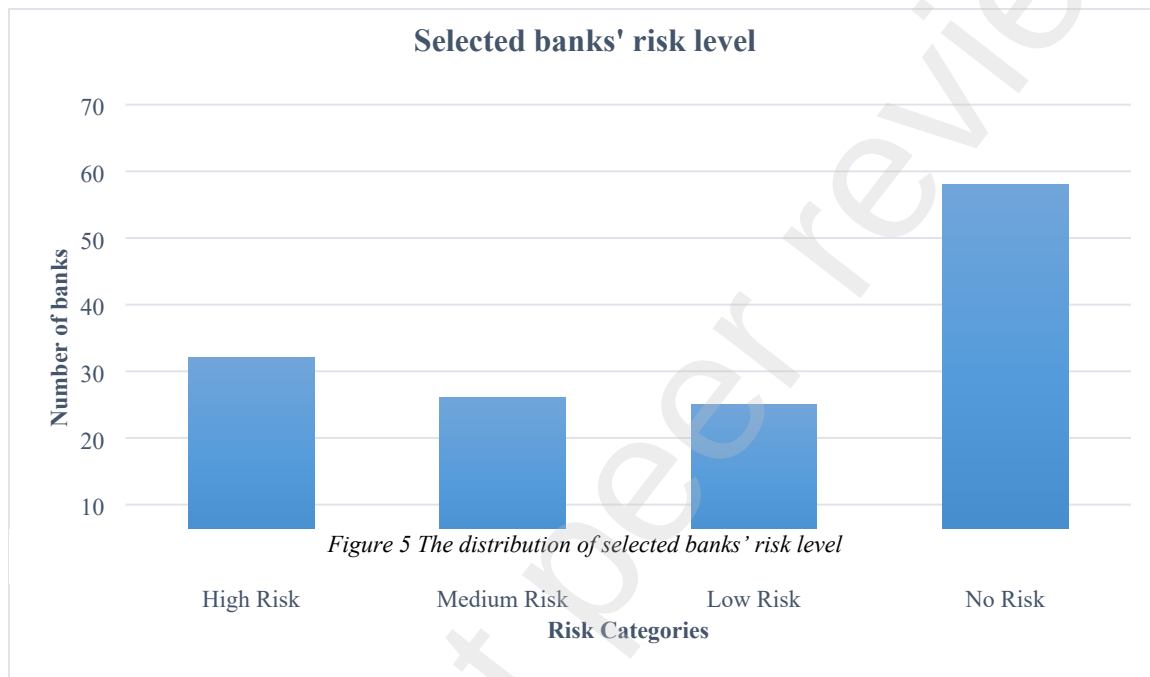


Figure 5 The distribution of selected banks' risk level

applying four proposed models. In order to evaluate the performance of models five metrics have been employed including: Accuracy, Precision, Recall, F1 score and Cross-Entropy Loss (CE Loss). They are obtained from confusion matrix which is defined as: a performance measurement for the machine learning classification problems where the output can be two or more classes. It is a table with combinations of predicted and actual values. It comprises four possible results. True positive (TP), True negative (TN), False positive (FP) and False negative (FN). The mentioned metrics determine the accuracy of models' classification and are calculated as follows:

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}} = \frac{TP + TN}{TP + TN + FP + FN} \quad (29)$$

$$Precision = \frac{TP}{TP + FP} \quad (30)$$

$$Recall = \frac{TP}{TP + FN} \quad (31)$$

$$F1 \text{ Score} = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (32)$$

$$CE \text{ Loss} = -[t \log(p) + (1 - t) \log(1 - p)] \quad (33)$$

Where t is the truth probability distribution of classification and p is the distribution of models' predicted probability. In this research, the figures for mentioned metrics are estimated for each model after classifying liquidity risk of banks. Then, by comparing the mentioned error figures related to four models, the best model for risk assessment of banks is identified, among the proposed approaches.

4.1. Regression

In the following, the most important variables which are able to affect liquidity risk more than other factors will be investigated by a regression method. As it was discussed earlier, 9 factors have been considered in this research including: Bank size, Capital ratio, Cost-To-Income ratio, ROA, ROE, GDP, Inflation rate, Unemployment rate and Interest rate. In order to understand selected data, figure 6 and 7 indicates the details of each variable which can be highly beneficial. They present the distribution of all variables' figures. Due to this information, we need to normalize the data to change them into a similar period. The data can be analyzed after normalization.

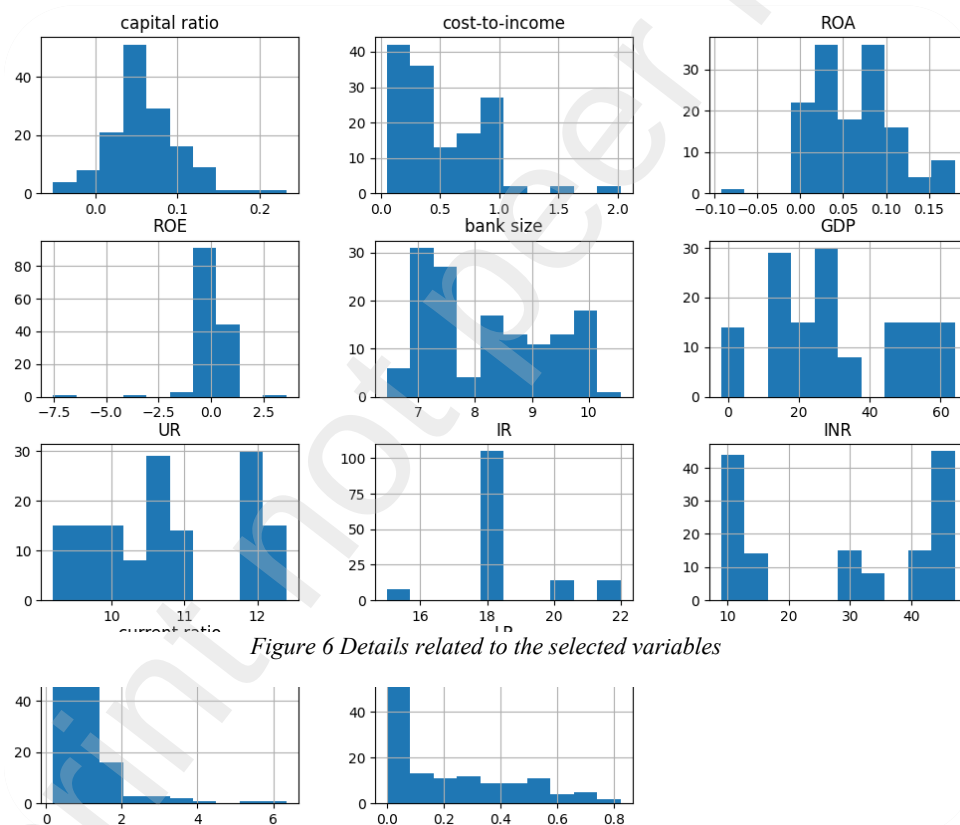


Figure 6 Details related to the selected variables

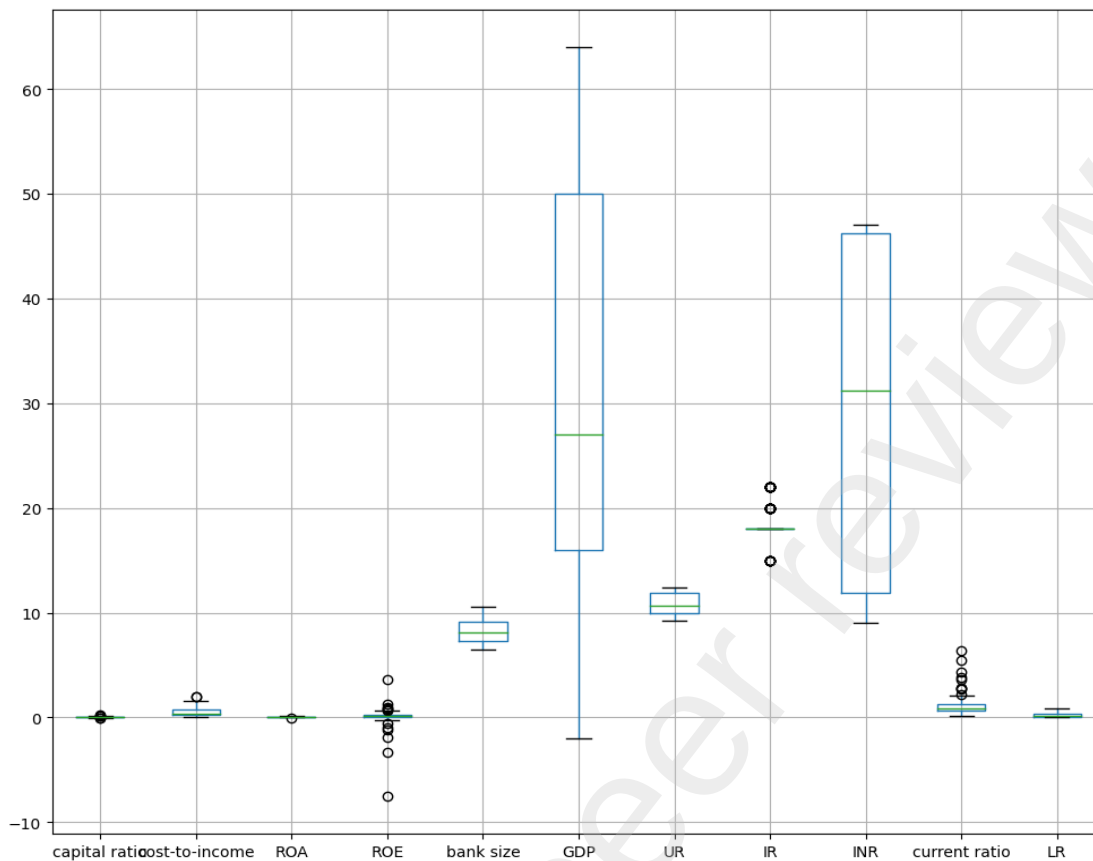


Figure 7 The distribution of variables' figures

We get informed of the amount of variables' importance by examining the coefficients which regression model has provided. Table 4 compares the coefficients of mentioned variables obtained from regression results.

Table 4 Regression results

Regression Results	Capital ratio	Cost-To-Income	ROA	ROE	Bank Size	GDP	UR	IR	INR
Coefficients	0.07	0.25	-0.12	0.09	0.23	-0.009	-0.03	-0.05	-0.03

It is obvious that the most important variables in liquidity risk of Iranian bank are Cost-To-Income ratio and the size of banks. These two variables have positive influence on liquidity. The most significant factor which has negative effect on liquidity risk is ROA. This issue can be highly beneficial for analysis related to banks. We can realize that if banks managers or authorities desire to control liquidity risk, they must consider the amount of costs and the specifications related to the bank size as well as the amount of return which is obtained from assets. It is worth noting that macroeconomic variables are not capable of affecting the banks' liquidity based on the regression results. They have insignificant negative effect on liquidity. Similarly, other two bank-specific factors, Capital ratio and ROE, have inconsiderable impression on banks' LR.

4.2. GSCV-DART-XGBOOST

In this section, we aim to present the specifications of models and their obtained results. The first model which will be examined is the GSCV-DART-XGBOOST. The mentioned hybrid model benefits from an optimization algorithm, named Grid Search-Cross Validation, which enables the model to obtain the best and the most suitable parameters in order to increase its accuracy. Moreover, classification methods can apply various algorithms in order to boost the data. Although most of the researches which employ XGBOOST use the default booster, this research intends to apply a more complex boosting algorithm which can lead to higher accuracy. As a result, Dropouts Meet Multiple Additive Regression Trees (DART) has been chosen for the hybrid model. Table 5 shows the details of chosen parameters which GSCV have suggested to apply. These measures have been selected testing thousands of possibilities resulting in highest accuracy.

Table 5 GSCV-DART-XGBOOST specifications

Model's Parameters	Objective	Max-Leaves	n-estimators	Max-Depth	Num-Class	Metric	Learning Rate	Subsample	Col-sample
GSCV-DART-XGBOOST	Multi-Softmax	10	100	2	4	Multi-Logloss	0.1	0.8	0.8

The GSCV-DART-XGBOOST hybrid model using mentioned measures can reach the lowest loss in the process of classification. In the following, the results of presented model will be illustrated by table 6. Therefore, the model's ability to classify the banks based on their liquidity risks can be realized.

Table 6 Evaluation metrics of GSCV-DART-XGBOOST

Model's Results	Accuracy	Precision	Recall	F1-Score	CE Loss
GSCV-DART-XGBOOST	0.972	0.974	0.972	0.971	0.08

As it is clear, the proposed hybrid model has been able to perform the classification, resulting in significant precision and considerably low loss. This hybrid method possesses high capability to be selected as the risk assessment model for banking system. Furthermore, figure 8 shows the number of real and predicted classifications obtained from proposed hybrid model. The GSCV-DART-XGBOOST predicted the banks classifications accurately and provided substantial results.

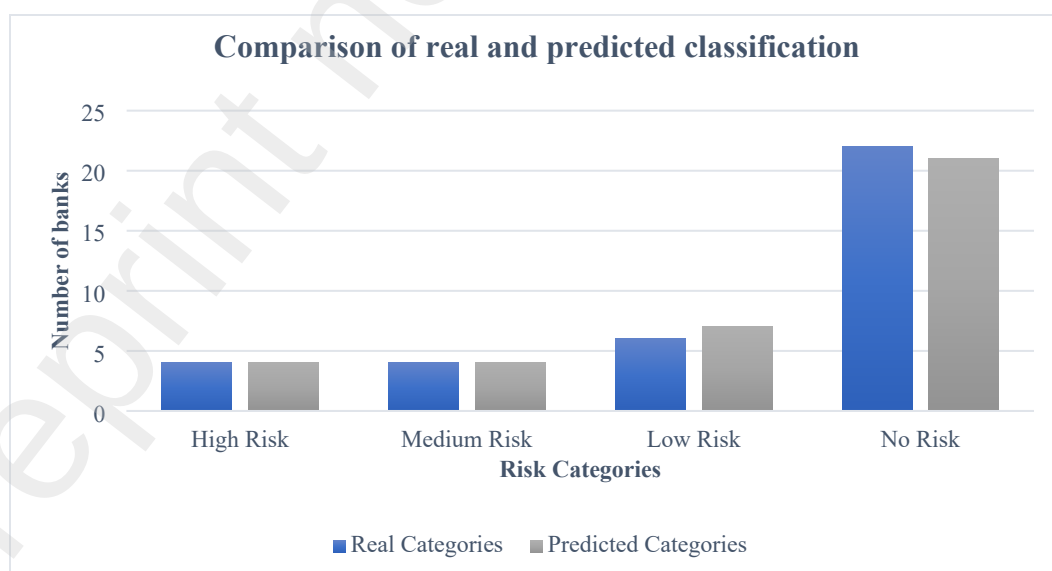


Figure 8 Comparison of real risk levels and predictions of GSCV-DART-XGBOOST

4.3. Bayesian-RF-LGBM

The second hybrid model which have been used in this research is Bayesian-RF-LGBM. There are different boosting algorithms which can perform in this model. These boosters are decisive factors in model's accuracy. As a result, we have decided to apply random forest booster in our chosen LGBM which demonstrated better results in experiments. On the other hand, the Bayesian Optimization Algorithm is crucial for selecting the best hyper-parameters of model and it was used to increase the accuracy and precision of our hybrid machine learning model. In the following, table 7 illustrates the most important parameters which have been suggested by Bayesian algorithm for training the hybrid model.

Table 7 Bayesian-RF-LGBM specifications

Model's Parameters	Objective	Num-Leaves	n-estimators	Max-Depth	Num-Class	Metric	Learning Rate	Subsample	Col-sample
Bayesian-RF LGBM	Multi-class	24	51	11	4	Multi-Logloss	0.065	0.8	0.8

The highest accuracy of model is obtained by applying mentioned measures. Thousands of combinations have been tested by Bayesian algorithm to reach the best parameters. As a result, the above combination of specifications is the best architecture of presented model for the proposed dataset classification. The model's findings illustrate substantial results as table 8 presents.

Table 8 Evaluation metrics of Bayesian-RF-LGBM

Model's Results	Accuracy	Precision	Recall	F1-Score	CE Loss
Bayesian-RF-LGBM	0.972	0.976	0.972	0.972	0.36

All 5 evaluations metrics confirm that the proposed hybrid model (Bayesian-Rf-LGBM) has great ability to be employed as a risk assessment system of banks' liquidity risk. For more analysis, figure 9 is presented to compare the number of predicted and real classifications. According to this figure, proposed hybrid model, Bayesian-RF-LGBM, had a substantial performance.

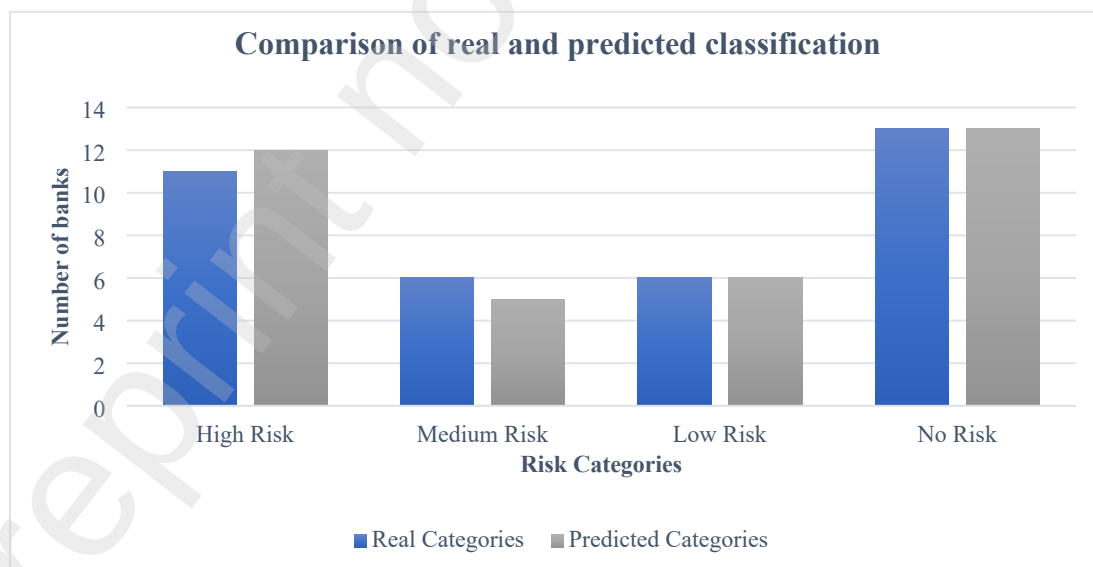


Figure 9 Comparison of real risk levels and predictions of Bayesian-RF-LGBM

4.4. LSTM

Long Short-Term Memory Networks is a deep learning, sequential neural network that allows information to persist. It is a special type of Recurrent Neural Network which is capable of handling the vanishing gradient problem faced by RNN. There are LSTM layers with tanh activation function which increase the ability of model in classification. We applied 3 LSTM layers and 4 Dense layers since it provides us with the highest accuracy. Table 9 explains the details of applied model.

Table 9 LSTM specifications

Layer	Output Shape	Function	Parameters
Zeropadding1D	(None, 11, 1)	-	0
LSTM	(None, 11, 64)	Activation = Tanh	16896
MaxPooling1D	(None, 5, 64)	-	0
LSTM	(None, 5, 64)	Activation = Tanh	33024
LSTM	(None, 16)	Activation = Tanh	5184
Dense1	(None, 64)	Activation = Tanh	1088
Dense 2	(None, 32)	Activation = Tanh	2080
Dense 3	(None, 4)	Activation = Softmax	132
Total parameters: 40212			
Trainable parameters: 40212			
Non-trainable parameters: 0			

The proposed model includes 40212 parameters which need to be trained. This illustrates the complexity of network. Following results have been obtained by applying LSTM model with mentioned specifications. Table 10 shows the capability of model for liquidity risk classification.

Table 10 Evaluation metrics of LSTM

Model's Results	Accuracy	Precision	Recall	F1-Score	CE Loss
LSTM	0.611	0.401	0.611	0.484	0.33

According to the findings, LSTM was not able to do an accurate classification as its metrics are not suitable enough. As a result, we can come to the conclusion that LSTM is not an appropriate method for our purpose and must not be employed by banks' authorities. For more understanding, figure 10 compares the test data for classification and the predicted ones. As it is clear, LSTM network could not predict well the categories of risks. Specially, LSTM was not able to determine the Low Risk and Medium Risk banks.

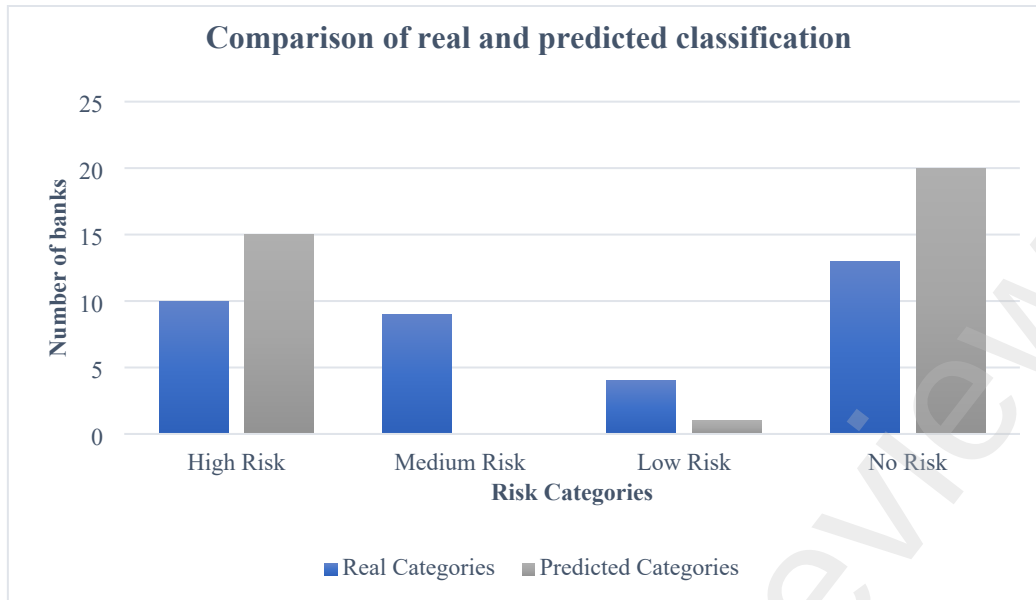


Figure 10 Comparison of real risk levels and predictions of LSTM

4.5. CNN

CNN is a substantial network with high ability of classifying. This network applies specific additional layers such as convolutional and pooling layers into model which results in more accuracy and enable model to have a deeper understanding of data. Our proposed CNN model is a complex network which consists of 6 CNN specific layers and 4 dense layers. Table 11 summarizes the structure of proposed CNN.

Table 11 CNN specifications

Layer	Output Shape	Function	Parameters
Zeropadding1D	(None, 11, 1)	-	0
Conv1D	(None, 11, 64)	Activation = Relu Padding = Same	384
Dropout	(None, 11, 64)	-	0
MaxPooling1D	(None, 10, 64)	-	0
Conv1D	(None, 10, 32)	Activation = Relu Padding = Same	10272
Dropout	(None, 10, 32)	-	0
MaxPooling1D	(None, 9, 32)	-	0
Conv1D	(None, 9, 16)	Activation = Relu Padding = Same	2576
Dropout	(None, 9, 16)	-	0
MaxPooling1D	(None, 8, 16)	-	0
Flatten	(None, 128)	-	0
Dense	(None, 128)	Activation = Relu	16512
Dense 1	(None, 64)	Activation = Relu	8256
Dense 2	(None, 32)	Activation = Relu	2080
Dense 3	(None, 4)	Activation = Softmax	132
Total parameters: 40212			
Trainable parameters: 40212			
Non-trainable parameters: 0			

The CNN model consists of 40212 parameters which were trained and this issue illustrates the complexity of proposed model. We have applied a CNN model with mentioned specifications and the results are presented in table 12. It can be comprehended that the proposed model was able to classify banks based on

their liquidity and its evaluation metrics were acceptable. However, two hybrid model outperformed this one.

Table 12 Evaluation metrics of CNN

Model's Results	Accuracy	Precision	Recall	F1-Score	CE Loss
CNN	0.833	0.836	0.833	0.824	0.10

Figure 11 is provided to introduce a comparison between real and predicted risk classifications of banks. Based on the figure, CNN model performed well for our intended classification but its accuracy and precision were lower compared to hybrid models.

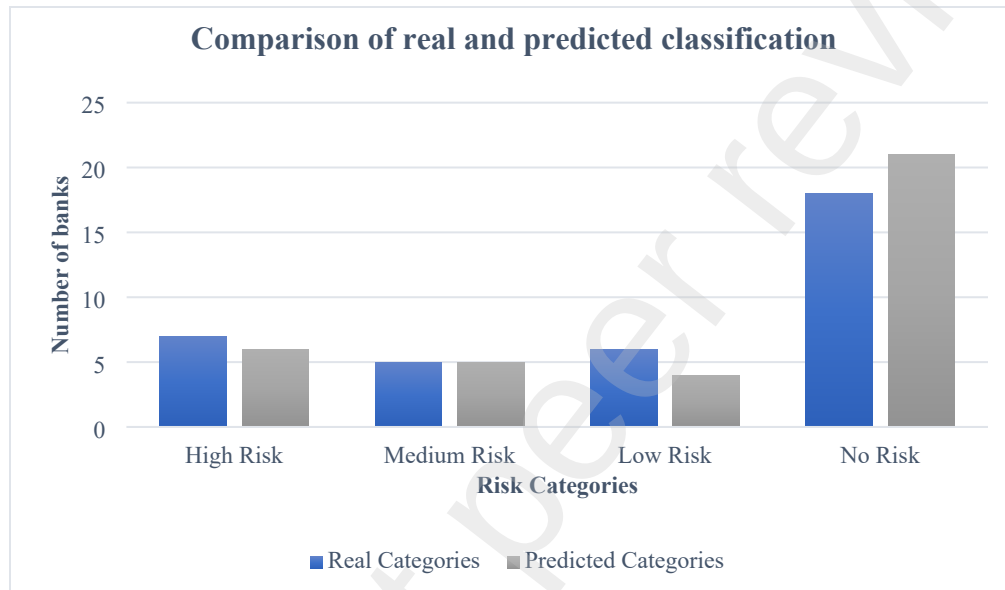


Figure 11 Comparison of real risk levels and predictions of CNN

4.6. Comparison of models

All proposed models have been employed and their results have been presented. Currently, their findings must be compared in order to find the best and the most accurate model for banks' liquidity risk assessment. We have evaluated the accuracy and the error of classifications by 5 metrics. Four of them illustrate the accuracy and one of them (CE Loss) shows the amount of error. Table 13 presents a comparison of four models' results. Based on the comparison of mentioned measures, it could be clear that all four proposed models had acceptable power for classification. However, two hybrid methods outperformed the other ones because they obtained higher accuracy measures.

Comparing two hybrid models, they had almost the same amount of accuracy based on the four measures (Accuracy, Precision, Recall, F1-Score). However, Considering the error metric, CE Loss, the GSCV-DART-XGBOOST model obtained a very low loss which indicates the high ability of this hybrid model for the research's purpose.

Table 13 Comparison of four proposed models' results

Model's Results	Accuracy	Precision	Recall	F1-Score	CE Loss
GSCV-DART-XGBOOST	0.972	0.974	0.972	0.971	0.08
Bayesian-RF-LGBM	0.972	0.976	0.972	0.972	0.36
LSTM	0.611	0.401	0.611	0.484	0.33

CNN	0.833	0.836	0.833	0.824	0.10
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Figure 12 summarizes the mis-predictions of four proposed models. It provides a great insight into the capability of methods. Consequently, two hybrid methods, GSCV-DART-XGBOOST and Bayesian-RF-LGBM, have the fewest number of false predictions and considering the amount of loss presented by table 13, GSCV-DART-XGBOOST is the most accurate model in order to apply for risk assessment system. It is worth noting that CNN network had an acceptable performance and LSTM was not a suitable method for our purpose due to its several false predictions.

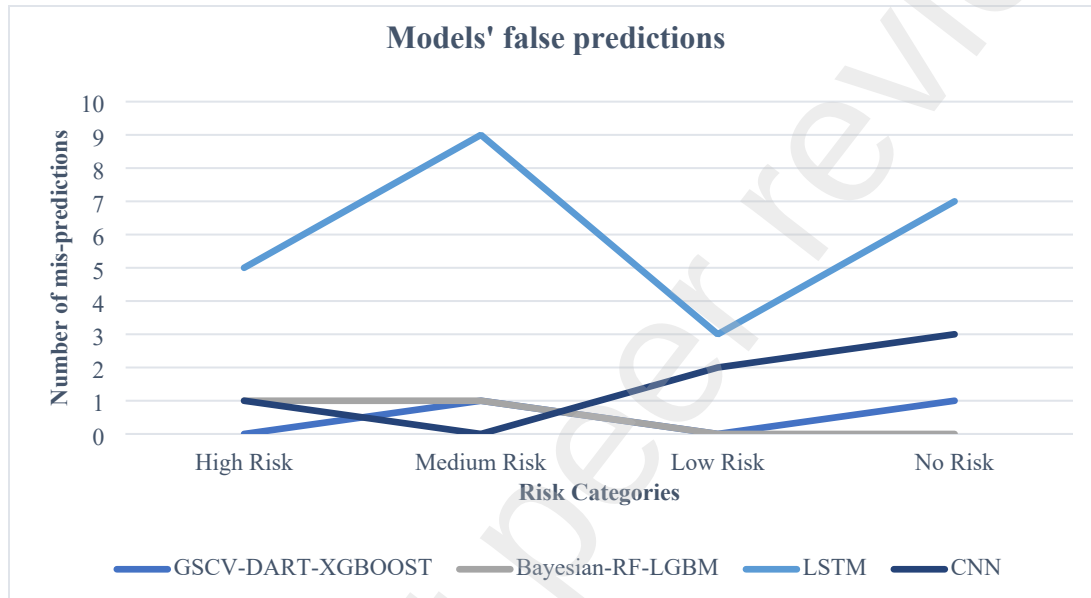


Figure 12 False predictions of four proposed models

5. Discussion

Bank liquidity risk is among the most vital risks threatening the profitability and survival of banks. Addressing this issue, both banks and their regulatory authorities are engaged in continuous monitoring of liquidity risk. This monitoring relies heavily on prominent banking industry measures, liquidity risk ratio and its significant determinants. As a result, selecting a reliable ratio and determining the most effective variables of liquidity risk can assist in presenting a risk assessment system in order to classify banks based on their risk level. Despite the importance of possessing an authentic risk assessment system, a limited number of researches have been carried out in this field. This research intends to take a big step in solving problems of bank liquidity risk by comparing machine learning and deep learning models to assess risk levels.

First, we did an in-depth study on the most related researches examining liquidity risk of banks. Their applied variables and LR ratios were summarized in table 1 and table 2. This study examined a total of 9 LR factors drawn from both scientific and literature sources affecting banks LR. Although macroeconomic variables affect significantly liquidity risk, some researches, Galleta & Mazzu (2019), Al_Duhaidahawi, et al (2020) and Tran (2020) for instance, have neglected these variables in their applied models and merely used important bank-specific factors which can reduce the models' accuracy.

This study introduced bank-specific and macroeconomic variables into liquidity risk models in order to boost their efficiency which is one of this article's benefits, similarly as Zhang & Zhao (2021) and Duan & Niu (2020). Four macroeconomic variables (GDP, Inflation, Unemployment, Interest rate) and five bank-specific factors (Size, Capital ratio, Cost-To-Income ratio, ROA and ROE) were analyzed in order to evaluate their influence on LR. Moreover, the authors decided to use current ratio as the proxy of LR. Providing a theoretical foundation for the selected LR variables and ratio can be a noticeable aspect of this research as well as including four important macroeconomic factors which can increase the accuracy of applied models.

The relationship between selected variables and liquidity risk has been evaluated by a regression model in order to assess the most significant factors of LR. The results illustrate that the most significant variables affecting liquidity risk of our selected banks positively are the size of banks and the Cost-To-Income ratio. Although Zaghdoudi & Hakimi (2017) and Choon, et al (2013) also found banks size as an effective variable, it was Zhang & Zhao (2021) who claimed that size of banks have positive influence on liquidity. With regard to Cost-To-Income, Ben Moussa (2015) concluded that banks expenses have no significant effect on liquidity which is in contrast to our results. On the other hand, Zhang & Zhao (2021) proved the substantial negative impact of costs on LR. The other important factor which this research has presented is ROA. The results show that ROA is related with LR negatively. Similarly, Jedidia & Hamza (2015) realized that ROA is a considerable factor of liquidity but with positive impact.

The research findings confirm that macroeconomic factors have insignificant impact on the amount of liquidity of banks. The results that Mazreku, et al (2019) and Trenca, et al (2015) acquired from their researches differ from the findings of this study because they concluded that macroeconomic variables are decisive elements of LR. However, Jia, et al (2020) confirm our findings as their research proves that macroeconomic variables are not important. It is worth mentioning that Trenca, et al (2015) and Choon, et al (2013) detected that ROE is an authentic variable affecting LR which this claim is not a confirmation of our findings since this study presents ROE as an inconsiderable factor.

After determining the most effective variables of liquidity risk, we aimed to apply four novel methods in order to classify banks into four risk categories based on their liquidity risk and compare their results to find the most accurate one. The selected models include two hybrid models and two deep learning methods including: GSCV-DART-XGBOOST, Bayesian-RF-LGBM, LSTM and CNN. One of the most important novelty of this research is applying four novel models, which in terms of the number of used models, can be known as prominent research. Similarly, presenting and applying two hybrid and optimized models can be considered as another novelty existing in this paper. The models which have been chosen have never been applied to present a risk assessment system in the literature of liquidity risk. As a result, the main novelty of this study is to present a risk assessment system using four novel methods including two hybrid ones which have been optimized in order to increase their accuracy.

As there is no studies which classified Iranian banks based on their liquidity risk level using machine learning methods, this study focuses on Iranian banks. The obtained data of Iranian banks from the beginning of 2013 to the end of 2022 have been applied for models' prediction. This study applied 5 evaluation metrics to compare the obtained results of models which all 5 are known as the best metrics. The Accuracy, Precision, Recall, F1 score and CE Loss criteria were calculated for the results of all four proposed models. Thus, the most accurate model determined in this research, have been approved by 5 metrics which this issue is another principal point of this research.

The Accuracy criterion of GSCV-DART-XGBOOST, Bayesian-RF-LGBM, LSTM and CNN were 0.972, 0.972, 0.611 and 0.833, respectively. Besides, The CE Loss of mentioned methods were 0.08, 0.36, 0.33

and 0.10, respectively. As it is clear, two hybrid models, GSCV-DART-XGBOOST and Bayesian-RF-LGBM, outperformed LSTM and CNN in this research, based on 5 metrics, and proved the ability of proposed hybrid methods. However, GSCV-DART-XGBOOST had lower loss compared to Bayesian-RF-LGBM. Consequently, GSCV-DART-XGBOOST is the most accurate model in this research and can be applied for risk assessment purposes.

The research findings illustrate that providing an in-depth analysis of banks which indicate their risk level have been successfully achieved. A reliable risk assessment system was presented by this research using GSCV-DART-XGBOOST model and it can be applied by various banks around the world, specially banks which comply with Iranian banking system regulations, to provide a substantial insight of the banking industry condition. The RAS can make a big difference in analyzing banks and improving their performance. Therefore, the results of this research are of significance importance and must be applied by banks.

5.1. Practical contribution

Research findings confirmed that GSCV-DART-XGBOOST model had the most accuracy in our risk assessment system. It classifies various banks into four categories based on their liquidity risk. This study can be highly beneficial in various countries. Due to the similarity of the banking system around the world, in terms of effective variables and managing liquidity risk, the proposed risk assessment system can also function accurately in several countries, specifically developing ones. Similarly, the most effective factors of liquidity risk have been assessed and analyzed in banking system. As a result, banks, around the globe, can obtain the ability to have more control on their liquidity risk level.

Diverse applications can be expected from the results of this research, especially for politicians and authorities. The banking system of different countries, will be able to benefit from the proposed risk assessment system, applying GSCV-DART-XGBOOST model, to monitor the liquidity risk condition of banks and examine their risk level. The banks managers and economic authorities, as well as organizations such as the Central Bank, Ministry of Economic, etc., can exploit the results of RAS to improve the banking system performance since the proposed RAS provide them with a remarkable perspective. This issue can solve a vast number of critical problems in the field of banking system since the results of applied policies during recent years can be analyzed. Moreover, as high-risk banks are prone to bankruptcy, the stated results can result in a marked decline of banks' failure. Due to the considerable effect of liquidity risk on the whole economy, the adverse results on other sectors of economy can be avoided.

5.2. Research proposes:

Based on the reviews and results, being aware of liquidity risk determinants and possessing a risk assessment system which provide a substantial insight of banking system in case of liquidity risk level, are of great importance in banking issues around the world, especially in Iran. In order to expand and develop the present research, proposes are presented as follows:

- Examining the effectiveness of other variables, policy environmental and financial structure factors for instance, can be highly beneficial to authorities and politicians and can provide them with great results.
- The future of liquidity risk can be forecasted by novel machine learning models if there is sufficient data. As it was proven in this research that machine learning models result in significant accuracy, they will be able to determine the future trend of liquidity risk of banks with lowest error. The findings of such research will have outstanding effect on the banking system efficiency.

- The application of proposed risk assessment system can be estimated in other banks and different condition to evaluate its accuracy.
- Comparing other novel models with this study's best model can confirm the results or present a new accurate model.

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